ROLL NO: _____

MADHAV INSTITUTE OF TECHNOLOGY & SCIENCE, GWALIOR

(A Deemed to be University | Autonomous Institute Established in 1957 | NAAC 'A++' Accredited)

END-SEM EXAMINATION — DECEMBER / JUNE 2024

B.TECH. ECE ENGINEERING — SEMESTER 4

Time Allowed: 3 Hours **Maximum Marks: 70**

NOTE: 1. Attempt any FIVE questions selecting at least ONE question from each Unit.
2. All questions carry equal marks. Sub-questions carry marks indicated in brackets [ ].
3. Assume suitable missing data, if any, and state it clearly.
4. Use of scientific non-programmable calculator is permitted.

--- UNIT I ---

Q.1 Classify Continuous and Discrete signals: Periodic, Energy, Power, Even, Odd signals. [14 Marks]

[OR]

Q.1 (Alt) Test Linear Time-Invariant (LTI) system properties: Linearity, Causality, Stability, Memory. [14 Marks]

--- UNIT II ---

Q.2 Evaluate Convolution Integral and Convolution Sum for LTI systems with graphical method. [14 Marks]

[OR]

Q.2 (Alt) Explain Fourier Series representation of continuous-time periodic signals and Dirichlet conditions. [14 Marks]

--- UNIT III ---

Q.3 Derive Continuous-Time Fourier Transform (CTFT) properties: Duality, Frequency shifting, Parseval's theorem. [14 Marks]

[OR]

Q.3 (Alt) State Nyquist Sampling Theorem. Explain Aliasing and reconstruction of signals using Sinc interpolation. [14 Marks]

--- UNIT IV ---

Q.4 Explain Laplace Transform ROC (Region of Convergence) properties. Find inverse Laplace transform. [14 Marks]

[OR]

Q.4 (Alt) Solve Differential Equations representing LTI systems using bilateral Laplace Transform. [14 Marks]

--- UNIT V ---

Q.5 Explain Z-Transform properties and ROC behavior. Find Z-transform of standard sequences. [14 Marks]

[OR]

Q.5 (Alt) Analyze Discrete LTI system stability and causality using Pole-Zero plots in Z-plane. [14 Marks]

**** END OF QUESTION PAPER — MITS GWALIOR AUTONOMOUS EXAMINATION CELL ****